

Choosing the Right Cloud Strategy

Your organisation must decide between **Public Cloud, Private Cloud, Hybrid Cloud, or Multi-Cloud** for its next-generation platform.

Based on what you learned in this module, discuss:

“Which cloud deployment strategy would you choose for a company handling both sensitive customer information and globally distributed high-traffic applications? Why?”

The Strategic Breakdown

1. Hybrid Cloud: Balancing Sensitive Data & Scalability

A hybrid cloud combines **Private Cloud** (or on-premises infrastructure) with **Public Cloud** services, allowing data and applications to be shared between them.

- **For Sensitive Customer Information:** You keep this data securely siloed in your **Private Cloud** environment. This gives the organization total control over data residency, governance, and strict regulatory compliance (like GDPR, HIPAA, or PCI-DSS).
- **For High-Traffic Applications:** You leverage the **Public Cloud** to host the customer-facing frontends and application logic. The public cloud offers virtually infinite scalability and can handle massive traffic spikes effortlessly.

2. Multi-Cloud: Achieving Global Distribution & High Availability

A multi-cloud strategy involves using cloud services from **multiple public cloud providers** (such as AWS, Google Cloud, and Microsoft Azure).

- **Global Distribution & Low Latency:** No single cloud provider has perfect coverage everywhere. By using multiple providers, you can deploy application nodes closer to your global user base, drastically reducing latency.
- **Redundancy and High Availability:** If one major cloud provider experiences a massive regional or global outage, your traffic can automatically failover to a completely independent secondary provider, ensuring your high-traffic application stays online.

Why the Other Options Fall Short

Cloud Strategy	Why it fails this specific scenario
Pure Public Cloud	While excellent for global traffic and scale, hosting highly sensitive customer data entirely on a shared public infrastructure introduces elevated compliance risks, potential regulatory penalties, and less granular control over data security.
Pure Private Cloud	While it offers maximum security for customer data, building and maintaining a private data infrastructure capable of handling massive, globally distributed traffic spikes is incredibly expensive, inefficient, and practically impossible to scale globally with low latency.